

Frederick Robinson

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Employment

- 2024–2026 **Principal Applied Scientist (later Member of Technical Staff), Amazon SF AI Lab**
- Early member of Amazon’s SF AI Lab, reestablishing infra and experimentation pipelines after Adept’s transition.
 - Founding member of the reinforcement learning group, setting up RL systems and advancing CoT/RLVR training.
 - Built [Turnstile \(Amazon Science\)](#), an open-source proxy capturing token-level rollout data for agentic RL training.
- 2024 **Member of Technical Staff, Adept**
- Improved reliability of computer-use agents by developing evaluation frameworks, data collection pipelines, and SFT experiments.
 - Contributed foundational work that supported the team’s transition into Amazon.
- 2022–2023 **Staff Machine Learning Engineer, Stripe, Inc. – Supportability Intelligence**
- Led an LLM-based enforcement initiative and a cross-functional team of ML and backend engineers, delivering > 20% improvement in Terms of Service violation detection.
 - Developed a transformer-based merchant website text classification model and a new service centralizing merchant business model data.
- 2017–2022 **Senior Applied Scientist, Amazon AWS – Interactive Video Service (Twitch)**
- Designed and productionized ML models to optimize data center selection for video playback, reducing global buffering rates by > 1%.
 - Built a streaming fraud detection system that improved viewbot detection by > 75%, and mentored applied scientists on ML research and engineering best practices.
- 2014–2017 **Software Engineer, Google, Inc. – Display Ads, Remarketing Quality**
- Built ML models and real-time bidding systems that optimized advertiser ROI and unlocked tens of millions in incremental annual revenue.

Skills

- ML Large Language Models, Reinforcement Learning (RLHF, CoT, RLVR), Supervised Fine-Tuning, Evaluation Frameworks, Distributed Training Systems, Data Collection Pipelines
- Tools PyTorch, slime, SGLang, Python, Golang

Education

- 2014 **Master of Arts in Mathematics, University of California Los Angeles**
- 2012 **Bachelor of Arts in Mathematics with Honors, Northwestern University**